Stability-Aware Reinforcement Learning for Autonomous Driving With Dynamics-Augmented State and Lyapunov Constraints

Yutao Luo, Andi Liu, Weiqiang Liang, and Jiawei Hong

Abstract—**Autonomous driving in extreme conditions presents substantial challenges in ensuring vehicle stability and safety. Traditional reinforcement learning (RL) methods for decision-making often lack vehicle dynamics modeling and formal stability constraints, leading to dynamically infeasible behaviors and unstable policy training. To overcome these limitations, we present an end-to-end reinforcement learning framework that incorporates a data-driven vehicle dynamics prediction model and Lyapunov-based stability constraints. The dynamics module is constructed using a hybrid Transformer architecture to effectively capture nonlinearities, time-varying parameters, and the coupling between longitudinal and lateral motions. This module captures the nonlinear interaction among vehicle, tire, and road, and provides predicted dynamic states as auxiliary inputs to enhance the RL state representation. Second, a neural network-based Lyapunov candidate function is incorporated into an enhanced Soft Actor–Critic (SAC) framework to impose stability-aware constraints on policy learning. To explicitly characterize lateral instability, the squared sideslip angle at the vehicle's center of gravity is defined as the Lyapunov cost. In addition, a hierarchical reward function is designed to balance multiple objectives during policy learning. The proposed framework is then validated through open-loop prediction experiments using both simulated and real vehicle data, followed by closed-loop evaluation in the CARLA simulator under representative driving scenarios, including low-friction road and dynamic obstacle avoidance. Experimental results show that the proposed method leads to significant improvements in both the stability and safety of the learned policies.**

Index Terms—**Dynamics, Reinforcement Learning, Constrained Motion Planning, Autonomous Driving.**

I. Introduction

AUTONOMOUS vehicles, as a core component of intelligent transportation systems, play a vital role in improving traffic efficiency and ensuring road safety [1], [2], [3]. Traditional autonomous driving systems are typically composed of four modules: perception, decision-making, motion planning, and feedback control. To guarantee safety and controllability in complex environments, vehicle dynamics models serve as a critical foundation for motion planning [4] and control [5]. These models predict the evolution of vehicle states in response to control commands and impose physical constraints to guarantee the dynamic feasibility and stability of the resulting trajectories.

In recent years, with the rapid development of deep learning techniques, autonomous driving systems have gradually evolved toward an end-to-end learning paradigm. This approach directly maps perception inputs to control commands or trajectory points via neural networks, significantly enhancing system adaptability and responsiveness. End-to-end learning methods mainly include imitation learning (IL) [6] and reinforcement learning (RL) [7], [8]. IL learns from expert demonstrations and offers high training efficiency and stability. However, it relies heavily on expert data and struggles to generalize under distribution shift in unseen environments. RL, in contrast, learns policies through trial-and-error interactions to maximize long-term cumulative rewards [9]. This enables RL to handle complex, high-dimensional, and dynamic decision-making tasks such as lane keeping, lane changing [10], merging [11], overtaking [12], and intersection handling [13], [14]. However, conventional RL frameworks suffer from training instability and often generate physically infeasible or unsafe actions due to the absence of vehicle dynamics priors, especially under extreme conditions such as low-friction surfaces and high-speed cornering. To address this problem, we incorporate vehicle dynamics priors and Lyapunov-based stability constraints into the RL framework to enhance decision-making under extreme scenarios.

Integrating vehicle dynamics models into reinforcement learning not only narrows the policy search space but also improves the robustness and safety of the system in complex environments. However, traditional physics-based models face challenges in accurately describing the nonlinear coupling between tire forces and road conditions, especially under low-friction road or emergency maneuvers. In addition, conventional methods struggle with high-dimensional continuous action spaces, and the need to solve complex differential equations in real time is often incompatible with the interactive nature of RL training. Building high-fidelity parametric models for each vehicle is also impractical due to the difficulty of parameter acquisition [15],

Received 8 May 2025; accepted 2 September 2025. Date of publication 25 September 2025; date of current version 9 October 2025. This article was recommended for publication by Associate Editor P. Borja and Editor C. D. Santina upon evaluation of the reviewers' comments. This work was supported in part by the Special Fund for High-Quality Development of Manufacturing Industry, MIIT, China, under Grant R-ZH-023-QT-001-20221009-001, in part by the Science and Technology Planning Project of Guangzhou, China, under Grant 2023B01J0016, and in part by the Guangdong Province 2024 Industrial Infrastructure Reconstruction Project Guangzhou Municipal Industry and Information Technology Bureau Letter [2024] No.60. *(Corresponding author: Yutao Luo.)*

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Digital Object Identifier 10.1109/LRA.2025.3615032

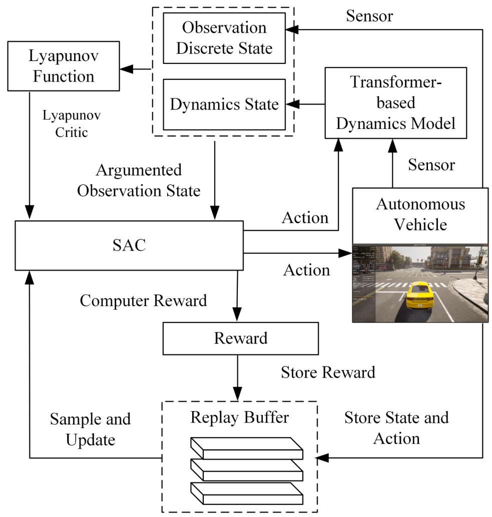

Fig. 1. Overall framework of the proposed decision-making system.

[16], model complexity, high computational cost, as well as limited generalization capability. To address these challenges, data-driven techniques have been adopted for vehicle kinematics and dynamics modeling by learning differentiable approximations from historical interaction data [17]. These methods can effectively capture complex nonlinear dynamics and support real-time prediction, making them more suitable for reinforcement learning. Prior studies have applied MLPs, RNNs, and LSTM networks for modeling vehicle dynamics [18], [19]. While these methods can capture nonlinear relationships between control inputs and vehicle states to some extent, they still face limitations in rapidly changing and multi-scale environments. Moreover, these models are typically confined to control modules, without being incorporated into full end-to-end decision-making frameworks.

To further improve model performance, we propose a hierarchical Transformer hybrid architecture. CNN layers are used to extract local physical features, while the Transformer captures global coupling of vehicle states through attention mechanisms. The predicted features are incorporated into the RL input to improve the agent's anticipation of future dynamic behavior. This enhances policy stability and control performance under extreme conditions such as low-friction surfaces, high-speed cornering, and obstacle avoidance. Additionally, we design a Lyapunov function approximated by a neural network. The sideslip angle, a key indicator of lateral tire force saturation, is used to enhance the Lyapunov function's responsiveness to lateral instability trends. The overall framework of the proposed model is illustrated in Fig. 1.

The main contributions of this letter are summarized as follows:

- A novel vehicle dynamics prediction model based on a hybrid Transformer architecture is proposed, which integrates local convolution and global attention to capture both short-term transients and long-term dependencies, and more effectively models nonlinear interactions under time-varying dynamics and longitudinal–lateral coupling.
- A stability-aware RL decision-making framework is developed, differing from conventional approaches by incorporating predicted dynamic states into policy inputs as physically meaningful priors. A Lyapunov-based constraint is further introduced during policy optimization to provide theoretical stability guarantees, where the squared sideslip angle is defined as a cost signal. This explicit constraint improves safety under high-risk scenarios and a multi-objective reward function is also designed, incorporating lateral acceleration as a key component.
- The proposed method is validated through open-loop prediction on real and simulated datasets, and closed-loop evaluation in an open-source simulator under extreme conditions such as low-friction road, high-speed cornering, and obstacle avoidance. Comparative results demonstrate the framework's effectiveness and robustness.

II. Vehicle Dynamics Model

A. Physics-Based Vehicle Dynamics Model

To balance the accuracy of vehicle motion characterization with computational efficiency, physics-based vehicle dynamics models are typically idealized. A three-degree-of-freedom (3-DOF) vehicle dynamics model is established to comprehensively capture the coupled relationships among longitudinal, lateral, and yaw motions.

$$\begin{cases} \dot{v}_x = v_y\, \omega_r + a_x, \\ \dot{v}_y = -\, v_x\, \omega_r + \frac{F_{yf}\cos\delta + F_{yr}}{m}, \\ \dot{\omega}_r = \frac{F_{yf}\, l_f - F_{yr}\, l_r}{I_z}, \end{cases} \tag{1}$$

where v_x and v_y are the longitudinal and lateral velocities; ω_r is the yaw rate; m is the vehicle mass; F_{yf} and F_{yr} are the lateral tire forces; δ is the front-wheel steering angle; l_f and l_r are the distances to the front and rear axles; a_x is the longitudinal acceleration; and I_z is the yaw moment of inertia.

At high driving speeds, the tire sideslip angle increases significantly, leading to pronounced nonlinear behavior in lateral tire forces. This nonlinearity is characterized by the saturation of lateral force as the sideslip angle increases. Moreover, load transfer effects during vehicle motion further influence the dynamic characteristics of the lateral forces, indicating a strong coupling between the longitudinal and lateral dynamics. Such coupling effects are difficult to model precisely using purely physics-based approaches. To address this, we adopt a data-driven approach and propose a hybrid Transformer architecture that integrates CNN-based local feature extraction with the temporal modeling capability of Transformers to improve prediction under coupled dynamics. To enhance learning efficiency, the model predicts the rate of change in state variables, i.e., Δv (m/s) and $\Delta\beta$ (rad), rather than absolute values, mitigating the limited temporal variation observed in short prediction intervals.

B. Transformer-Based Vehicle Dynamics Model

This section introduces the proposed Transformer-based vehicle dynamics model, along with its architecture, data preprocessing, and training setup. As illustrated in Fig. 2,

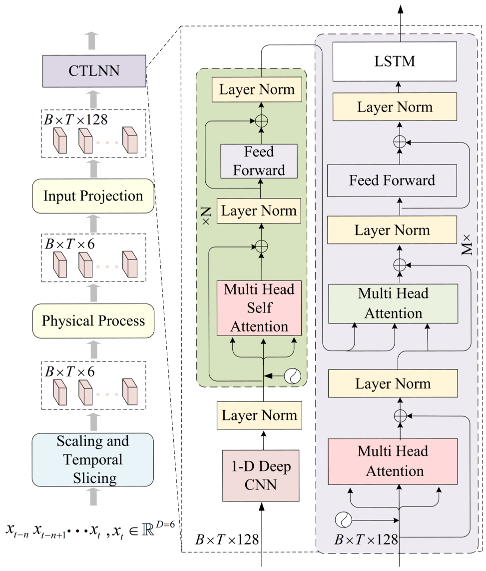

Fig. 2. Illustration of the proposed Transformer hybrid network for vehicle dynamics prediction.

TABLE I
CONFIGURATION OF THE PROPOSED VEHICLE DYNAMICS MODEL

Parameter	**Value**
Sequence length	10
LSTM layers	2
Hidden units per LSTM layer	128
Transformer encoder layers	2
Attention heads	8
Transformer hidden dimension	128
Transformer decoder head count	4
Neurons of CNN hidden layers	[16, 32, 64, 128]
Learning rate	1×10^{-4}
Batch size	1024

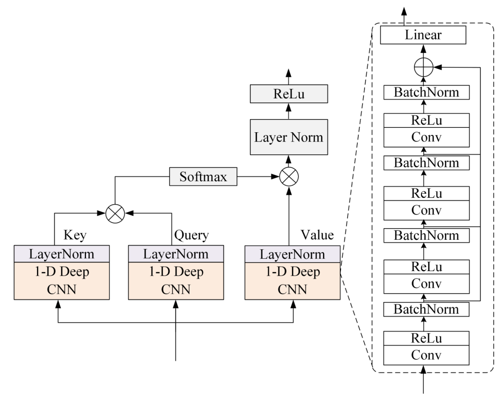

Fig. 3. The local–global attention module integrates 1D-CNN-based local encoding of dynamics with dot-product attention to capture temporal dependencies.

the input comprises a continuous sequence of vehicle dynamic states and control commands, denoted as $\mathbf{X} = [\mathbf{x}_t, \mathbf{x}_{t-1}, \ldots, \mathbf{x}_{t-n+1}, \mathbf{x}_{t-n}]$, where each $\mathbf{x}_t \in \mathbb{R}^D$ encodes six key variables including the longitudinal velocity v_x, lateral velocity v_y, yaw rate ω_r, steering angle δ, throttle, and brake, forming a feature vector of dimension $D = 6$. The Scaling and Temporal Slicing module normalizes the raw inputs and segments them into fixed-length subsequences of shape $[B, T, D]$, where B is the batch size and T is the sequence length. The Physical Process module simulates the vehicle's physical response to control inputs and reflects the real evolution of vehicle dynamics. This module is not involved in neural network training; instead, it is used to generate target state variables for the training data. Based on the bicycle model, we define the vehicle speed $v = \sqrt{v_x^2 + v_y^2}$ and the sideslip angle $\beta = \arctan(v_y/v_x)$. The sideslip angle is an important indicator of lateral stability, while vehicle speed reflects the overall motion state of the vehicle. This hybrid design integrates physical knowledge with data-driven modeling, enhancing model interpretability and mitigating the limited generalization of analytical models under complex driving scenarios. Then, the input sequence is passed through a linear projection layer that maps the input into a higher-dimensional latent space $\mathbb{R}^{B \times T \times H}$, where $H = 128$ denotes the hidden feature dimension used throughout the CTLNN architecture. This projected representation is then fed into the CTLNN module for temporal feature extraction and state prediction. The architectural and training configuration of the proposed model is summarized in Table I.

To enhance the modeling capacity of vehicle dynamic response, our architecture incorporates a local temporal encoder using stacked one-dimensional convolutional layers (1D-CNN) and a global dependency module via Transformer attention. The 1D-CNN module captures local temporal dynamics such as actuator delays and sensor noise, modeled as high-frequency components. Specifically, the module applies three layers of temporal convolution with kernel size 3, operating along the time axis. The number of channels increases progressively through the layers, allowing the network to model increasingly abstract local temporal features. In addition, the Transformer module enables the modeling of long-range temporal dependencies and interactions. Through global attention, it learns non-local causality in historical states, such as the gradual build-up of sideslip angle on low-friction roads, or stability degradation due to sustained high-speed driving. The encoder employs a multi-head self-attention mechanism to capture temporal dependencies among state variables and control commands. The decoder uses a multi-head cross-attention mechanism to associate the encoded features with historical target dynamic states, allowing vehicle dynamic features across different time steps to interact and influence each other. The temporal input spans several seconds, which is sufficient to capture critical dynamic effects such as sideslip accumulation, yaw instability, and coupled throttle and steering interactions. This design enhances the model's ability to capture trends in state evolution.

As shown in Fig. 3, to capture the long-term dependencies in temporal features, the output sequence obtained from the local feature extraction module is first projected through three linear layers to produce the query matrices Q, key matrices K, and value matrices V for subsequent attention computation. Multi-head attention (MHA) computes the correlations between features in parallel through multiple attention heads. Its computation is formulated as

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V, \tag{2}$$

where d_k denotes the dimensionality of the key vectors. The high-level representations with dynamic structure obtained through the global attention mechanism are ultimately fed into the LSTM module to generate sequence predictions of future vehicle dynamic states, including key indicators such as sideslip angle β and vehicle speed v. The implementation code will be made publicly available at: https://github.com/aaaaaaaann/LRSAC.

III. Lyapunov-Constrained Reinforcement Learning

A. Data-Based Stability Analysis

The autonomous driving decision-making problem can be formulated as a model-free Markov Decision Process (MDP), represented by a five-tuple $(\mathcal{S}, \mathcal{A}, P, R, \gamma)$, where $\mathcal{S}$ and $\mathcal{A}$ denote the state space and action space. The transition of the state is modeled by the transition probability $P(s_{t+1} \mid s_t, a_t)$. R is the immediate reward and $\gamma \in [0, 1)$ is the discount factor. A cost function $c(s_t, a_t)$ is used to evaluate the quality of a state-action pair. In stabilization tasks, the cost function is typically defined as $c(s_t, a_t) = \mathbb{E}_p \|s_{t+1}\|$, aiming to minimize the state norm over time. The cost function under the policy π is defined as $c_\pi(s_t) \triangleq \mathbb{E}_{a\sim\pi}\, c(s_t, a_t)$.

Definition 1: (Uniformly Ultimately Bounded, UUB).([20]). A system is considered uniformly ultimately bounded (UUB) with respect to $c_\pi(\cdot)$, provided that there exist positive constants η, b; for all $\epsilon < b$, there exists $T(\epsilon, \eta)$ such that $c_\pi(s_1) \le \epsilon \Rightarrow c_\pi(s_t) \le \eta, \forall\, t \ge T$.

To analyze stability, we adopt a Lyapunov-based approach. This involves constructing a positive definite Lyapunov function that decreases along system trajectories, guiding the state toward regions of lower energy or cost. This principle underpins the guarantee of UUB, where the system remains within a cost-bounded safe region.

Theorem 1: ([21]). Assume there exists a function $L : \mathcal{S} \to \mathbb{R}_+$, along with positive constants $\alpha_1, \alpha_2, \eta, \alpha_3$, such that the following inequalities are satisfied:

$$\alpha_1\, c_\pi(s) \le L(s) \le \alpha_2\, c_\pi(s), \quad \forall s \in \mathcal{S}, \tag{3}$$

$$\mathbb{E}_{s\sim\mu_\pi}[\mathbb{E}_{s'\sim\mathcal{P}}[L(s') - L(s)]] \le -\,\alpha_3\, \mathbb{E}_{s\sim\mu_\pi}[c_\pi(s)]\,, \tag{4}$$

where

$$\mu_\pi(s) \triangleq \lim_{N\to\infty} \frac{1}{N} \sum_{t=0}^{N-1} \mathbb{P}\big(s_t = s \,\big|\, \rho, \pi, t\big)\,, \tag{5}$$

is the (infinite) sampling distribution. N is the maximum instant that the probability of being in the edge set is greater than zero, i.e., $N = \max\{\, t : \mathbb{P}(s \in \Delta \mid \rho, \pi, t) > 0 \,\}$. Then, the system is UUB stable with ultimate bound η; the expectation $\mathbb{E}_{p(s|\rho,\pi,t)} c_\pi(s)$ is bounded during the N time steps.

B. Lyapunov Soft Actor–Critic Algorithm

The Soft Actor–Critic (SAC) algorithm is a representative method of maximum entropy reinforcement learning. By introducing an entropy regularization term into the objective function, SAC achieves a better balance between exploration and exploitation. The policy entropy $\mathcal{H}\big(\pi(\cdot \mid s_t)\big)$ is defined as:

$$\mathcal{H}\big(\pi(\cdot \mid s_t)\big) = -\,\mathbb{E}_{a_t\sim\pi} \log \pi(a_t \mid s_t). \tag{6}$$

We propose formulating a Lyapunov critic that depends on both the state s and the action a, and satisfies the condition $L(s) = \mathbb{E}_{a\sim\pi}\, L_c(s, a)$, where $L(s)$ denotes the Lyapunov function used to evaluate the stability condition in Theorem 1.

$$L_c(s, a) = f_\phi(s, a)^\top f_\phi(s, a), \tag{7}$$

where f_ϕ is the output of a fully connected neural network with parameter ϕ. It ensures $L_c(s, a)$ remains positive definite, and $L_c(s, a)$ is updated to minimize the objective function,

$$J(L_c) = \mathbb{E}_{(s,a)\sim\mathcal{D}}\Big[\frac{1}{2}\big(L_c(s, a) - L_{\text{target}}(s, a)\big)^2\Big], \tag{8}$$

where L_{target} is the approximation target for L_c. $\mathcal{D}$ is the set of collected transition pairs. The approximation target is

$$L_{\text{target}}(s, a) = c(s, a) + \gamma\, L_c'(s', a'), \tag{9}$$

where L_c' is the target network parameterized by ϕ', updated through exponentially moving average of weights of L_c. In our implementation, the cost function $c(s, a)$ is defined as the square of the vehicle's sideslip angle, a quantity that reflects lateral dynamic deviation with physical interpretability. In the Lyapunov difference condition, $c(s, a)$ functions as a weighting term that modulates the expected descent rate of the Lyapunov function. By Theorem 1, this construction enables the Lyapunov expectation to decrease proportionally to the cost, satisfying the UUB under finite-horizon sampling, as shown in Fig. 4.

In this subsection, the policy learning problem is summarized as the following constrained optimization problem,

$$\begin{aligned} &\text{find } \pi_\theta \\ &\text{s.t.} \;\; \mathbb{E}_\mathcal{D}[L_c(s', f_\theta(\epsilon, s')) - L_c(s, a) + \alpha_3 c(s, a)] \le 0, \\ &\quad\;\; -\,\mathbb{E}_\mathcal{D}[\log \pi(a \mid s)] \ge \mathcal{H}_t, \end{aligned} \tag{10}$$

where the stochastic policy π_θ is parameterized by a neural network f_θ, the policy objective function is formulated as:

$$\begin{aligned} J(\pi_\theta) = \mathbb{E}_{(s,a,s',r)\sim\mathcal{D}} &\left[\alpha \log \pi_\theta(f_\theta(\epsilon, s) \mid s) - Q(s, f_\theta(\epsilon, s))\right. \\ &+ \lambda\, (L_c(s', f_\theta(\epsilon, s')) - L_c(s, a) \\ &\left. + \alpha_3 c(s, a)) + \nu\, (L_c(s, a) - e)\right]. \end{aligned} \tag{11}$$

The Q-function critic is trained using the standard temporal-difference (TD) objective from the SAC framework:

$$\begin{aligned} J(Q) = \mathbb{E}_{(s,a,r,s')\sim\mathcal{D}} \frac{1}{2}\, &[r(s, a) + \gamma\, \mathbb{E}_{a'\sim\pi}[Q'(s', f_\theta(\epsilon, s')) \\ &-\alpha(\log \pi_\theta(f_\theta(\epsilon, s') \mid s'))] - Q(s, a)]^2 \end{aligned} \tag{12}$$

The values of Lagrange multipliers α and λ are adjusted by gradient ascent to maximize the following objectives respectively while being clipped to be positive,

$$\begin{aligned} J(\alpha) &= \alpha\, \mathbb{E}_{\mathcal{D}}[\log \pi_\theta(a \mid s) + \mathcal{H}_t]\,, \\ J(\lambda) &= \lambda\, \mathbb{E}_{\mathcal{D}}[L_c(s', f_\theta(\varepsilon, s')) - L_c(s,a) + \alpha_3 c]\,. \end{aligned} \tag{13}$$

During training, the Lagrange multipliers are updated by

$$\begin{aligned} \alpha &\leftarrow \max\big(0,\, \alpha + \delta\, \nabla_\alpha J(\alpha)\big), \\ \lambda &\leftarrow \max\big(0,\, \lambda + \delta\, \nabla_\lambda J(\lambda)\big), \end{aligned} \tag{14}$$

where δ is the learning rate.

Traditional autonomous driving frameworks typically define the state space using kinematic features of the ego and surrounding vehicles, such as longitudinal velocity, heading angle, and relative position to a reference path. While effective under nominal conditions, such representations often fail to generalize in complex scenarios, low-adhesion roads, high-speed cornering, or emergency maneuvers, due to oversimplified vehicle dynamics that limit the generation of physically feasible actions. To address this, we augment the state space with dynamics-consistent variables, including the vehicle speed, to better capture coupled longitudinal and lateral dynamics and assess potential instability or rollover risks. On this basis, we design a multi-objective reward function with interpretable terms, where the weights are heuristically tuned following prior reinforcement learning studies [7]. A collision penalty $R_c = -10$ and a lane departure penalty $R_o = -1$ discourage unsafe behaviors, while a steering penalty $R_s = -|\delta|^2$ suppresses abrupt control. Driving efficiency is encouraged by a speed reward R_v, dynamically adjusted by the deviation from the target velocity. Additionally, a trajectory deviation penalty R_d is applied in real time by identifying the vehicle's distance to the nearest reference waypoint, guiding it to follow the planned path. To avoid the agent falling into a stagnant state, a low-speed penalty $R_l = -0.8$ is set. Finally, to improve dynamic stability under extreme conditions, a lateral acceleration penalty $R_a = 0.01 \cdot a_y$ is incorporated to constrain instability risks on slippery or sharply curved roads.

IV. Open-Loop Testing

A. Simulation Dataset

The simulation dataset was collected using the CARLA platform, enabling controlled reproduction of diverse driving scenarios and precise acquisition of vehicle dynamics data for model training. Key state and control variables including lateral and longitudinal velocities, yaw rate, steering angle, throttle, and brake were continuously recorded to capture the vehicle's dynamic responses and control actions. Data were gathered from various road geometries, including straight paths, curves, and urban intersections, where the vehicle executed maneuvers such as lane changes, obstacle avoidance, and U-turns. To ensure generalization across surface conditions, the dataset includes sequences on both dry and wet roads with reduced friction, supporting robust learning under different adhesion regimes.

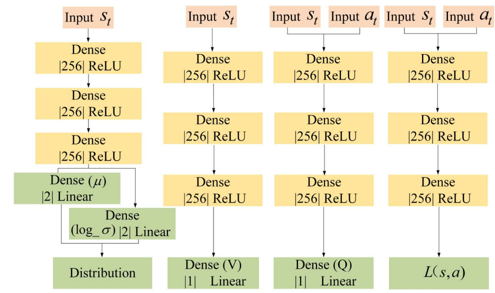

Fig. 4. Network architecture of the Lyapunov-based RL, including the policy, value, Q-value, and Lyapunov critic networks. Each module is composed of three hidden layers.

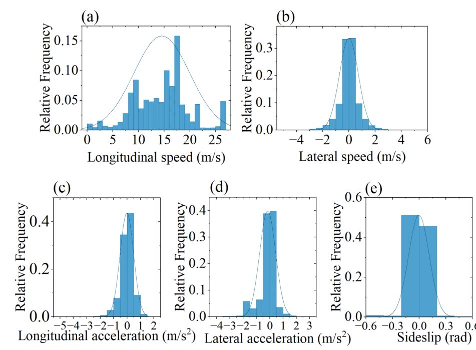

Fig. 5. Distributions of vehicle dynamic states. (a) Longitudinal speed. (b) Lateral speed. (c) Longitudinal acceleration. (d) Lateral acceleration. (e) Sideslip.

TABLE II
Physical Parameters of the Test Vehicle

Parameter	Value
Vehicle mass (kg)	2136.6
Center of gravity height (unladen) (m)	0.49
Yaw moment of inertia ($\mathrm{kg \cdot m^2}$)	2914.3
Track width (m)	1.62
Wheelbase (m)	2.9

B. Real-World Dataset

To validate the practical applicability of the learned dynamics models, we additionally collected a real-world dataset comprising over 100,000 samples from naturalistic driving using a domestic production vehicle. The data span diverse road surfaces, driving behaviors, and environmental conditions, including ambient temperature variations. The acquisition platform integrates modules for environmental perception, GNSS and inertial localization, decision and control, and actuator communication. High-frequency CAN bus logging captures both control commands and vehicle dynamics. As shown in Fig. 5, the dataset exhibits rich variability across velocity, acceleration, and sideslip angle. Key physical parameters of the test vehicle, including mass, wheelbase, track width, center of mass height, and yaw inertia, are listed in Table II.

TABLE III
MODEL PERFORMANCE COMPARISON ON THE SIMULATION DATASET

Model	Params (K)	MAE		RMSE	
		Δv (m/s)	$\Delta\beta$ (rad)	Δv (m/s)	$\Delta\beta$ (rad)
CNN	889.21	0.073	0.0086	0.10	0.032
Attn+LSTM	925.05	0.052	0.0076	0.071	0.035
GRU	595.58	0.084	0.010	0.12	0.046
LSTM	661.63	0.086	0.011	0.11	0.043
Transformer	799.10	0.063	0.0084	0.089	0.034
Neural ODE	701.01	0.23	0.016	0.36	0.061
Physics-based	–	0.43	0.062	0.56	0.090
Ours	683.92	**0.045**	**0.0070**	**0.067**	**0.025**

TABLE IV
MODEL PERFORMANCE ON THE REAL-WORLD DATASET

Model	MAE		RMSE	
	Δv (m/s)	$\Delta\beta$ (rad)	Δv (m/s)	$\Delta\beta$ (rad)
CNN	0.023	0.0082	0.026	0.011
Attn+LSTM	0.0089	0.0026	0.013	0.0037
GRU	0.0054	0.0021	0.0099	0.0031
LSTM	0.0049	0.0015	0.0096	0.0029
Transformer	0.0048	0.0020	0.0090	0.0036
Neural ODE	0.015	0.0047	0.023	0.0057
Physics-based	0.10	0.012	0.18	0.017
Ours	**0.0047**	**0.0012**	**0.0072**	**0.0015**

TABLE V
EFFECT OF HIDDEN DIMENSION AND LAYER NUMBER ON MODEL PERFORMANCE

Hidden-dimension	Layers	MAE		RMSE	
		Δv (m/s)	$\Delta\beta$ (rad)	Δv (m/s)	$\Delta\beta$ (rad)
64	1	0.061	0.0076	0.079	0.029
128	1	0.057	0.0074	0.080	0.029
256	1	0.054	0.011	0.085	0.043
512	1	0.067	0.0080	0.084	0.037
128	2	**0.045**	**0.0070**	**0.067**	**0.025**
256	2	0.052	0.0083	0.081	0.040
256	3	0.051	0.0085	0.071	0.039

TABLE VI
ABLATION STUDY OF TEMPORAL WINDOW LENGTH

T	MAE			RMSE		
	Δv (m/s)	$\Delta\beta$ (rad)	Total	Δv (m/s)	$\Delta\beta$ (rad)	Total
1	0.13	0.010	0.070	0.17	0.038	0.104
5	0.053	**0.0059**	0.029	0.069	**0.022**	**0.045**
10	**0.045**	0.0070	**0.026**	**0.067**	0.025	0.046
15	0.050	0.0088	0.029	0.072	0.039	0.056
20	0.053	0.0089	0.031	0.076	0.036	0.056

TABLE VII
ABLATION STUDY OF MODEL COMPONENTS

Model	MAE		RMSE	
	Δv (m/s)	$\Delta\beta$ (rad)	Δv (m/s)	$\Delta\beta$ (rad)
w/o 1D-CNN	0.079	0.0073	0.11	0.035
w/o Transformer	0.060	0.0078	0.079	0.032
w/o LSTM	0.068	0.010	0.095	0.031
w/o Transformer & 1D-CNN	0.080	0.0082	0.10	0.039
w/o Transformer & LSTM	0.13	0.013	0.16	0.044
w/o LSTM & 1D-CNN	0.099	0.0091	0.12	0.040
Full Model (Ours)	**0.045**	**0.0070**	**0.067**	**0.025**

C. Open-Loop Prediction Results

We evaluate our proposed vehicle dynamics prediction model against several baselines, including CNN, GRU, LSTM, Transformer, Neural ODE, and a physics-based approach. As shown in Table III, our model achieves the lowest errors. Specifically, compared to Transformer and LSTM models, our approach reduces the higher mean absolute error (MAE) of Δv by 28.6% and 47.7%, the MAE of $\Delta\beta$ by 16.7% and 36.4%, demonstrating its superior ability to capture nonlinear and coupled dynamics.While attention-enhanced LSTM (Attn+LSTM) slightly outperforms the vanilla LSTM but still lacks sufficient global temporal modeling. GRU and CNN show moderate performance but exhibit higher prediction errors in complex motion scenarios. Neural ODE, though continuous in nature, suffers from error accumulation, leading to the MAE or root-mean-square error (RMSE), which limits its reliability in safety-critical tasks. The physics-based model, despite its interpretability, yields the worst performance overall, particularly in sideslip angle prediction, indicating its limited adaptability under complex conditions. To further assess generalization, we conduct open-loop prediction on real-world vehicle data collected from naturalistic driving; as shown in Table IV, our model maintains the lowest errors under complex urban scenes and variable road surfaces, consistently outperforming both sequence- and physics-based baselines and demonstrating strong applicability to field deployment.

As shown in Table VII, removing any single component from the model architecture CNN, LSTM, or Transformer leads to increased prediction errors, confirming their complementary roles in capturing multi-scale temporal dependencies. Transformer-only models outperform LSTM-only ones in long-horizon tasks, while CNN enhances short-term feature extraction when combined with either. As shown in Table V, Table VI, for parameter configuration, a two-layer LSTM with 128 hidden units achieves the best trade-off between accuracy and complexity; increasing size or depth beyond this yields marginal or negative gains. In terms of temporal input, extending the sequence length from 1 to 10 significantly improves performance, particularly for sideslip angle prediction, while longer windows introduce noise and slightly degrade accuracy. Overall, the results highlight that accurate vehicle dynamics prediction relies on combining local and global temporal modeling, along with careful tuning of model capacity and input horizon.

V. CLOSED-LOOP TESTING

Simulation experiments were conducted on the CARLA platform using the Town03 map, which features urban roads, narrow alleyways, and sharp turns that pose significant challenges for autonomous driving. To increase task complexity, dynamic obstacles were randomly introduced into the training environment.

To evaluate the proposed algorithm's stability and robustness under strongly coupled dynamics, three representative test scenarios were designed in the CARLA environment (Figs. 6 and 7). Map-1 features a complex curved road on a low-friction road, with continuous S-curves, U-turns, and frequent steering, making lateral control particularly challenging. Map-2 is a closed-loop circular road that simulates real-world roundabouts

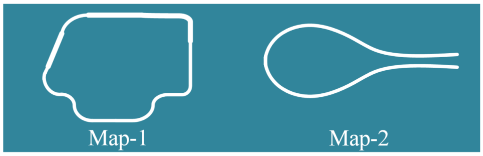

Fig. 6. Layouts of test tracks with extreme driving conditions in simulation.

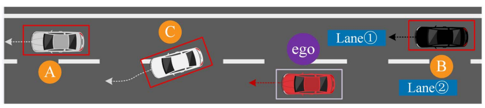

Fig. 7. Emergency lane-change interference scenario (The ego vehicle (purple box) faces sudden interference from vehicle C changing lanes. Vehicles A and B represent surrounding traffic.).

and highway ramps. The third scenario involves emergency obstacle avoidance, where dynamic obstacles suddenly change lanes to interfere with the ego vehicle during driving. These scenarios encompass extreme conditions such as aggressive maneuvers and nonlinear dynamics, effectively highlighting the proposed method's advantages in maintaining stability and ensuring safety.

In order to verify the effectiveness of the proposed method for enhancing vehicle stability, the models in this letter are compared with several other models, including the traditional SAC decision model, the RSAC model, which adds a lateral acceleration reward on top of the augmented state space, and the proposed LRSAC model, which incorporates vehicle dynamics-based stability constraints.

A. Evaluation Metrics

We evaluate task performance using three standard metrics: success rate, collision rate, and timeout rate, reflecting route completion, safety, and efficiency. To assess stability, we report four dynamic variables: sideslip angle, speed, lateral acceleration, and yaw rate, capturing the amplitude, variability, and stability of vehicle. In emergency obstacle avoidance, we further adopt mean trajectory curvature to quantify path smoothness and feasibility. The point-wise curvature is computed as:

$$\kappa(t) = \frac{|\dot{x}(t)\ddot{y}(t) - \dot{y}(t)\ddot{x}(t)|}{(\dot{x}(t)^2 + \dot{y}(t)^2)^{3/2}}, \tag{15}$$

where $\dot{x}(t)$, $\dot{y}(t)$ and $\ddot{x}(t)$, $\ddot{y}(t)$ are first and second derivatives of position. The average $\bar{\kappa} = \frac{1}{N}\sum_{i=1}^{N} \kappa_i$ is used as the final smoothness metric.

B. Closed-Loop Testing Results

As shown in Table VIII, SAC exhibits significant instability in lateral dynamics. The maximum sideslip angle reaches 0.46 rad, and the peak lateral acceleration is 13.31 m/s^2 with an average of 2.87 m/s^2. These sharp dynamic fluctuations indicate a high risk of oversteering and tail instability during cornering. RSAC mitigates some of the abrupt lateral Table VIII responses through penalty terms, reducing the lateral acceleration peak to 9.58 m/s^2 and slightly lowers the MAE to 2.76 m/s^2. However, it introduces greater variability in sideslip, suggesting incomplete coordination of stability objectives. In contrast, LRSAC further improves stability by limiting the sideslip peak to 0.33 rad, a 28.3% reduction compared to RSAC, and reducing the lateral acceleration MAE to 1.93 m/s^2, a 30% decrease. In addition, LRSAC achieves lower speed variance and smoother yaw rate transitions, indicating superior control through sharp curves. According to Table X, LRSAC achieves a 100% success rate in Map-1, outperforming RSAC's 96%, and further obtains a significantly higher average test reward of 105.76.

As shown in Table IX, in the circular track scenario, SAC suffers further degradation in control performance on the circular road, the sideslip angle fluctuates widely with a maximum of 0.22 rad and a minimum of -0.27 rad, while lateral acceleration peaks at 10.46 m/s^2. RSAC improves attitude stability by penalizing lateral acceleration, reducing yaw rate variability and limiting lateral force magnitude. However, this comes at the cost of maneuverability, as its average speed drops to 7.36 m/s. LRSAC achieves the best overall balance: it maintains the highest average speed, the lowest MAE and standard deviation of sideslip angle, and smoother lateral acceleration. Its yaw rate remains responsive, yet well-regulated. These results demonstrate that

TABLE VIII
DYNAMIC PERFORMANCE COMPARISON ON MAP-1

Classification	Model	Max	Min	MAE	Std
Sideslip (rad)	SAC	0.46	-0.29	**0.080**	0.12
	RSAC	0.51	-0.36	0.098	0.14
	LRSAC	**0.33**	**-0.28**	0.088	**0.11**
Speed (m/s)	SAC	8.11	0	5.77	**1.48**
	RSAC	**13.37**	0	**6.14**	2.55
	LRSAC	8.46	0	5.62	1.92
Lateral Acceleration (m/s^2)	SAC	13.31	-11.10	2.87	4.10
	RSAC	**9.58**	-9.05	2.76	3.52
	LRSAC	9.99	**-8.42**	**1.93**	**2.71**
Yaw rate (deg/s)	SAC	109.96	-85.29	22.58	33.23
	RSAC	62.89	-76.71	22.69	28.89
	LRSAC	87.82	-73.84	20.68	27.18

TABLE IX
DYNAMIC PERFORMANCE COMPARISON ON MAP-2

Classification	Model	Max	Min	MAE	Std
Sideslip (rad)	SAC	0.22	-0.27	0.081	0.088
	RSAC	0.14	-0.16	0.047	0.057
	LRSAC	**0.11**	**-0.13**	**0.039**	**0.048**
Speed (m/s)	SAC	8.96	0	3.91	**1.63**
	RSAC	9.32	0	7.36	1.84
	LRSAC	**10.05**	0	**8.07**	2.08
Lateral Acceleration (m/s^2)	SAC	10.46	-23.06	2.49	3.45
	RSAC	**8.53**	**-8.16**	2.66	**3.35**
	LRSAC	10.95	-8.62	**2.65**	3.54
Yaw rate (deg/s)	SAC	31.37	-53.94	12.77	14.63
	RSAC	42.39	-59.11	18.48	22.28
	LRSAC	58.51	-55.55	19.11	23.67

TABLE X
TASK COMPLETION STATISTICS IN MAP-1 AND MAP-2

Map	Model	Success rate(%)	Crash rate(%)	Overtime rate(%)	Average Test rewards
Map-1	LRSAC	**100**	**0**	**0**	**105.76**
	RSAC	96	4	0	75.96
Map-2	LRSAC	**100**	**0**	**0**	**199.20**
	RSAC	92	8	0	138.02

TABLE XI
TRAJECTORY SMOOTHNESS AND LATERAL STABILITY IN EMERGENCY AVOIDANCE

Model	Mean curvature	Lateral Acceleration RMS
SAC	0.0308	4.98
RSAC	0.0407	2.94
LRSAC	**0.0269**	**2.69**

TABLE XII
DYNAMIC PERFORMANCE IN EMERGENCY AVOIDANCE

Classification	Model	Max	Min	MAE	Std
Sideslip (rad)	SAC	**0.088**	-0.14	0.029	0.040
	RSAC	0.13	-0.079	0.026	0.036
	LRSAC	0.12	**-0.054**	**0.021**	**0.035**
Speed (m/s)	SAC	11.78	0	7.93	3.55
	RSAC	10.28	0	5.28	**2.58**
	LRSAC	**12.30**	0	**8.63**	3.70
Lateral Acceleration (m/s^2)	SAC	14.27	-12.15	3.75	4.99
	RSAC	**6.23**	**-6.32**	**2.18**	**2.70**
	LRSAC	7.57	-7.52	2.31	2.95
Yaw rate (deg/s)	SAC	62.86	-55.93	13.51	19.70
	RSAC	34.36	-28.69	7.26	10.35
	LRSAC	37.86	-28.68	9.80	13.38

LRSAC enables both efficient and stable turning under sustained curvature, confirming the positive impact of Lyapunov-based constraints on system dynamics. According to Table X, LRSAC achieves a 100% success rate on Map-2, outperforming RSAC's 92%, and obtains a higher average reward.

In emergency avoidance scenarios, the vehicle must respond rapidly to dynamic obstacles while maintaining stability. As shown in Tables XI and XII, the SAC policy results in unstable trajectories with a high mean curvature of 0.0308, and large sideslip angle fluctuations, and peak lateral acceleration reaching 14.27 m/s^2. The yaw rate spans over 118.8 deg/s, indicating abrupt steering and poor control precision. RSAC improves smoothness by reducing lateral acceleration and narrowing sideslip variation; however, this comes at the expense of responsiveness, with average speed dropping to 5.28 m/s. In contrast, LRSAC performs best: it generates the smoothest trajectory (lowest mean curvature), minimizes sideslip variation, and maintains a low lateral acceleration MAE, and reduces the Root Mean Square (RMS) of lateral acceleration to 2.69, indicating improved lateral motion smoothness. Its yaw rate remains responsive yet stable. The average speed is sustained at 8.63 m/s, demonstrating a superior balance between agility, precision, and dynamic safety.

VI. CONCLUSION

We proposed a stability-aware reinforcement learning framework integrating a Transformer-based vehicle dynamics predictor with Lyapunov-based constraints. The neural predictor augments the policy input with dynamics-consistent priors, while the Lyapunov critic shapes the policy to satisfy a descent condition associated with sideslip-based costs. Open-loop experiments on simulated and real data demonstrate higher prediction accuracy than sequence- and physics-based baselines. Closed-loop evaluations in CARLA confirm improved lateral stability, smoother trajectories, and higher task success rates under low-friction and interference. Future work includes real-time deployment and cross-platform generalization.

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